

WALTON-ON-THE-NAZE, United Kingdom

WMO: 036960

Lat: 51.851N

Lon: 1.267E

Elev: 16

StdP: 14.69

Time Zone: 0.00 (EUW)

Period: 94-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.6	31.5	24.0	18.2	34.1	26.0	20.0	35.3	36.7	46.7	32.7	44.8	10.5	300	0.625

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.4	74.0	66.0	72.0	65.1	70.5	64.1	67.9	72.4	66.6	70.7	65.3	69.4	9.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	95.7	71.0	64.7	91.7	69.9	63.5	87.7	68.6	32.2	72.6	31.2	70.7	30.2	69.5	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.8	26.5	23.2	25.6	80.1	0.9	2.5	24.9	81.9	24.4	83.4	23.8	84.8	23.2	86.6		
(5)			24.8	70.6	0.8	1.5	24.3	71.7	23.8	72.5	23.4	73.4	22.8	74.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.8	41.3	41.7	44.2	48.2	53.9	59.5	63.7	64.4	60.2	54.4	47.2	42.3
	DBStd	9.48	4.68	4.92	4.78	4.43	3.99	4.20	3.89	3.84	3.93	4.63	5.01	5.33
	HDD50	1153	270	233	186	84	12	0	0	0	0	16	109	243
	HDD65	4890	734	652	644	503	343	169	70	59	149	329	535	703
	CDD50	1820	1	1	7	31	134	287	426	446	307	152	24	4
	CDD65	81	0	0	0	0	0	6	31	39	5	0	0	0
	CDH74	65	0	0	0	0	0	8	23	32	2	0	0	0
	CDH80	1	0	0	0	0	0	0	0	1	0	0	0	0
Wind	WSAvg	11.5	13.9	13.0	11.9	10.8	11.0	10.2	9.6	9.4	10.7	12.8	11.9	13.2
Precipitation	PrecAvg	22.3	2.0	1.6	1.3	1.5	1.7	2.0	1.8	2.1	1.9	2.4	2.3	2.1
	PrecMax	29.5	4.2	3.7	3.5	3.9	3.8	4.8	3.7	4.5	3.9	5.8	4.6	4.1
	PrecMin	15.3	0.4	0.2	0.4	0.0	0.3	0.4	0.4	0.2	0.3	0.0	0.7	0.5
	PrecStd	3.7	0.9	0.9	0.8	1.0	1.0	1.3	0.8	1.3	1.0	1.3	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	54.6	59.1	62.2	67.5	74.9	76.9	77.4	72.7	65.9	59.8	55.5
		MCWB	50.3	49.6	51.5	54.4	60.5	64.5	66.6	67.2	64.0	59.7	56.4	52.4
	2%	DB	51.8	52.1	55.4	58.7	63.9	70.3	73.6	74.3	69.7	63.4	57.4	53.2
		MCWB	48.5	48.4	49.7	52.8	57.8	61.9	65.2	67.1	63.3	58.9	54.3	50.8
	5%	DB	49.6	50.2	52.9	56.5	61.9	67.8	71.8	72.4	67.8	62.0	55.6	51.7
		MCWB	47.1	46.6	48.4	51.7	56.2	60.7	64.7	66.0	62.5	58.4	52.7	49.6
	10%	DB	47.9	48.5	51.0	54.4	60.2	66.0	70.2	70.8	66.3	60.9	53.9	50.0
		MCWB	45.6	45.5	47.3	50.2	55.1	60.0	63.8	64.9	61.3	57.8	51.5	47.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.1	51.1	52.9	55.8	61.4	66.1	68.7	70.2	66.2	61.5	57.3	53.1
		MCDB	53.2	53.4	57.0	59.8	65.1	71.7	74.3	74.8	69.8	63.6	59.2	54.7
	2%	WB	49.1	48.9	50.7	53.9	58.7	63.7	67.3	68.6	64.7	60.3	54.8	51.3
		MCDB	51.1	51.2	54.1	57.7	63.0	68.2	72.2	72.7	68.1	62.2	56.5	52.7
	5%	WB	47.5	47.4	49.3	52.2	57.1	62.3	66.0	67.3	63.7	59.4	53.2	49.7
		MCDB	49.3	49.5	52.2	55.6	61.0	66.4	70.4	71.2	66.9	61.3	54.8	51.3
	10%	WB	45.8	46.0	47.8	50.8	55.7	61.1	64.7	66.1	62.4	58.4	51.9	48.2
		MCDB	47.6	48.0	50.3	54.0	59.3	65.2	69.0	70.0	65.6	60.5	53.7	49.8
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.9	8.7	9.4	9.4	10.2	10.2	10.4	9.7	8.6	7.8	6.8
		MCDBR	9.0	9.4	11.2	12.2	11.4	12.3	11.9	11.4	10.8	8.9	8.1	7.6
		MCWBR	8.2	7.4	7.6	7.2	6.9	7.6	6.9	6.7	7.2	6.7	6.8	7.1
	5% WB	MCDBR	8.9	8.9	10.1	10.6	10.3	10.4	9.9	9.6	9.3	7.5	7.3	7.4
		MCWBR	8.5	7.6	7.5	6.8	7.1	7.7	6.6	6.5	6.9	6.5	6.9	7.1
Clear-Sky Solar Irradiance	taub	0.331	0.348	0.384	0.405	0.411	0.414	0.412	0.403	0.382	0.365	0.337	0.321	
	taud	2.374	2.381	2.277	2.249	2.278	2.294	2.312	2.358	2.407	2.428	2.414	2.365	
	Ebn at Noon	212	243	254	263	267	267	266	261	254	233	208	194	
	Edh at Noon	20	25	34	39	40	40	38	35	30	24	19	17	
All-Sky Solar Radiation	RadAvg	242	455	809	1291	1545	1689	1607	1344	1008	581	295	188	
	RadStd	23	68	97	126	113	131	151	101	81	54	33	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page